

Table 1: Popular programming platforms for RFID engineering and antenna design

| Category                        | Type               | Tool/Platform                         | Key role in RFID/ antenna design              |
|---------------------------------|--------------------|---------------------------------------|-----------------------------------------------|
| EM (electromagnetic) simulation | 3D EM solver       | ANSYS HFSS                            | High-accuracy antenna and RFID tag simulation |
| EM simulation                   | EM solver          | CST Studio Suite                      | Time/frequency domain antenna optimisation    |
| EM simulation                   | EM solver          | FEKO                                  | Large-scale RFID system simulation            |
| EM simulation                   | Multiphysics       | COMSOL Multiphysics                   | RF + thermal coupled analysis                 |
| EM simulation                   | Time-domain solver | XFDTD                                 | Radiation and SAR analysis                    |
| EM simulation                   | Open source EM     | OpenEMS                               | Low-cost antenna prototyping                  |
| EM simulation                   | Planar EM          | Sonnet Suites                         | Microstrip RFID antenna design                |
| EM simulation                   | 3D EM tool         | EMPro                                 | EM + circuit co-simulation                    |
| EM simulation                   | Lightweight EM     | NEC (Numerical Electromagnetics Code) | Wire antenna modelling                        |
| Circuit design                  | RF circuit tool    | ADS (Advanced Design System)          | RF front-end and co-simulation                |
| Circuit design                  | Circuit simulator  | LTspice                               | Analog RFID circuit design                    |
| Circuit design                  | IC design          | Cadence Virtuoso                      | RFID chip/ASIC development                    |
| Circuit design                  | Circuit simulator  | PSpice                                | SPICE-based RF simulation                     |
| Circuit design                  | Circuit tool       | QUCS                                  | RF circuit prototyping                        |
| System tools                    | System control     | LabVIEW                               | RFID testing and automation                   |
| System tools                    | Network simulator  | OMNeT++                               | RFID protocol simulation                      |
| System tools                    | Network simulator  | NS-3                                  | Wireless/RFID modelling                       |
| System tools                    | Security tool      | RFIDIOT                               | RFID protocol testing                         |
| PCB design                      | PCB tool           | Altium Designer                       | Professional RFID hardware design             |
| PCB design                      | PCB tool           | KiCad                                 | Open source PCB prototyping                   |
| PCB design                      | PCB tool           | Autodesk Eagle                        | Rapid prototyping                             |
| PCB design                      | PCB tool           | OrCAD PCB Designer                    | Industry-grade PCB layout                     |
| PCB design                      | PCB + simulation   | Proteus                               | Embedded RFID system design                   |



openEMS is a free and open electromagnetic field solver using the FDTD method.

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**Welcome! openEMS is a free and open electromagnetic field solver using the FDTD method. Matlab or Octave and Python are used as an easy and flexible scripting interface.**

## Install

[Latest Windows Build](#)

[Linux Build instructions](#)

## News and Announcements

**01.01.2023:** Since the forum is still down, I have activated a github discussion to ask your question about the usage of openEMS or just give some feedback. [Github Discussions](#)

**07.12.2022:** New temporary openEMS.de website since the [University DUE](#) hosting is currently offline!

Figure 1: OpenEMS